

①. Class Person

Data Members:

name
age

Method

Set Person (name, age)

this.name = name

this.age = age

End class

Class Student Inherits Person

Data members:

regNo
marks[5]
average

Method:

Set Student (regNo, marks)

this.regNo = regNo

this.marks = marks

Calculate Average()

sum = 0

for i = 0 to

sum = sum + mark[i]

End for

average = sum / 5

End class

Class ResearchStudent Inherits Student

Data Members:

Research Area

Method:

Set Research Area (area)

Research Area = area

displayCompleteDetails()

Display "Name:", name

Display "Age:", age

Display "Register No.:", regNo

Display "Marks:", marks

Display "Average:", average

Display "Research Area:", researchArea.

End class.

Main Program

Create object rs of Research student

rs.setPerson("Rohit", 22)

rs.setStudent("23CS901", {85, 90, 88, 92, 95})

rs.calculateAverage()

rs.setResearchArea("Artificial Intelligence")

rs.displayCompleteDetails()

End

(2)

Begin

Read N

Read Array A

For $i = 0$ to $N-1$

For $j = i+1$ to $N-1$

If $A[i] < A[j]$

$N = N-1$

$j = j-1$

End If

End For

Print Modified Array

End

②. Class Area
 Method area(length)
 Return length * length
 End Method.
 Method Area (length, breadth)
 Return length * breadth
 End Method.
 End Class.

③. Trace the execution.

Initial Sum = 0

(i)	Even / odd	Operation	Sum
1	odd	0 - 1	-1
2	even	-1 + 2	1
3	odd	1 - 3	-2
4	even	-2 + 4	2
5	odd	2 - 5	-3

$$\boxed{O/p = -3}$$

⑤. Begin
 Read N
 Read Array A
 Largest $\leftarrow -\infty$
 Second Largest $\leftarrow -\infty$
 For $i \in 0$ to $n-1$
 if $A[i] > \text{largest}$ then
 Second largest $\leftarrow \text{largest}$
 largest $\leftarrow A[i]$
 Else if $A[i] > \text{Secondlargest}$
 And $A[i] \neq \text{largest}$ then
 Second largest $\leftarrow A[i]$
 End if
End for
Print Secondlargest
End.

⑥. Class Animal
 Method speak ()
 Print "Animal"
 end Method
 end class
 class Dog inherits Animal
 Method speak ()
 Print "Barks"
 end Method
 end class
 Animal Obj $\leftarrow$ new dog ()
 Obj.speak ()

③. Begin

Interface Payment

Method pay (amount)

End interface

Class CreditCard Implements Payment

Method pay (amount)

Print "Paid", amount, "using Credit Card"

End Method

End class

Class UPI Implements Payment

Method pay (amount)

Print "Paid", amount, "using UPI"

End Method.

End Class.

Class Netbanking Implements Payment

Method pay (amount)

Print "Paid", amount, "using Net Banking"

End Method.

End class.

Read choice

Read amount

If choice = 1

payment ← Credit Card

Else if choice = 2

payment ← UPI

Also

payment ← Net Banking

End if

payment.pay (amount)

End.

⑨.

Class Employee

DATA

id .

name

Constructor (id, name)

 this.id ← id

 this.name ← name.

end constructor

end class .

9)

Begin

Class Vehicle

Data

Vehicle No

ownerName

Method display()

Print vehicle No

Print ownerName

End Method.

End Class.

Class Car inherits Vehicle

Data

FuelType

Method display()

Print Vehicle No

Print ownerName

Print fuelType

End Method

End Class

Class Truck inherits Vehicle

Data

Load Capacity

Method display()

Print vehicle No

Print ownerName

Print Load Capacity

End Method

End Class

Read N

For i ← 1 to N

 Read type

 If type = "car" Then

 Create Car obj

 Read VehicleNo, ownerName, fuelType

 Store Obj

 Else ~~Else~~

 Create Truck obj

 Read Vehicle No, ownerName, fuelType

 Store Obj

 End if ~~End if~~

End for

For Each obj In Vehicle List

 obj.display()

End For

End.